UNIVERSIDAD POLITÉCNICA DE SAN LUIS POTOSÍ

INGENIERÍA EN TECNOLOGÍAS DE LA INFORMACIÓN

ACTIVIDAD 06:

Implementación IPSec VPN

Universidad Politécnica de San Luis Potosí

Materia:

CNO V Seguridad Informática

Estudiante:

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Fecha de entrega:

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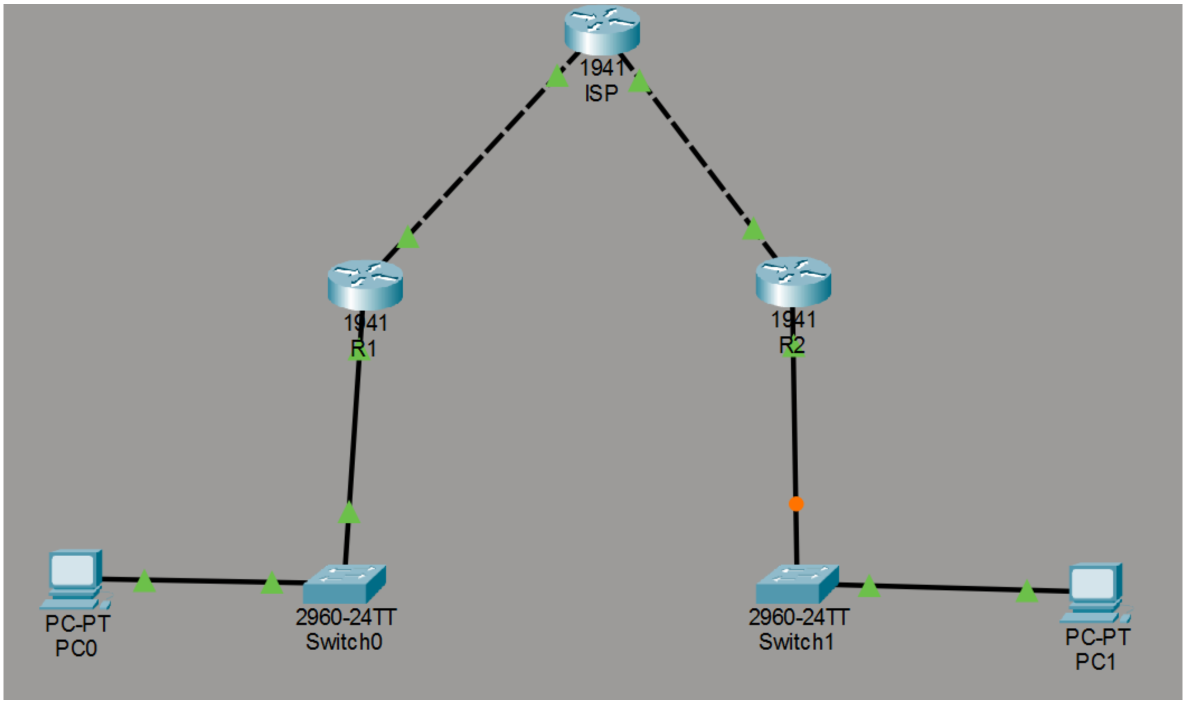
Topología



- Tres routers 1941 (ISP, R1, R2)
- Dos switches 2960
- Dos clientes (PC´s)

1.Configuracion Inicial

Se asignan direcciones IP a cada router y se activan las interfaces. También se agrega una ruta para que R1 y R3 puedan salir hacia el ISP y comunicarse entre sí por internet.

Configuración R1

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostame R1
                   ^
% Invalid input detected at '^' marker.

Router(config)#hostname R1
R1(config)#interface g0/1
R1(config-if)#ip address 192.168.1.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R1(config-if)#interface g0/0
R1(config-if)#ip address 209.165.100.1 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

R1(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.100.2
R1(config)#
```

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Configuración R2

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface g0/1
R2(config-if)#ip address 192.168.3.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

R2(config-if)#interface g0/0
R2(config-if)#ip address 209.165.200.1 255.255.255.0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

R2(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.2
R2(config)#
```

Configuración ISP

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname ISP
ISP(config)#interface g0/0
ISP(config-if)#ip address 209.165.100.2 255.255.255.0
ISP(config-if)#no shutdown

ISP(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

ISP(config-if)#interface g0/1
ISP(config-if)#ip address 209.165.200.2 255.255.255.0
ISP(config-if)#no shutdown

ISP(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

ISP(config-if)#ip route 0.0.0.0 0.0.0.0 209.165.200.1
ISP(config)#
```

2.Licencia de seguridad habilitada

Se habilita el paquete de seguridad del router. Sin esta licencia, el router no permite usar funciones VPN/IPsec.

Licencia R1

```
R1(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module name = C1900 Next reboot level =
securityk9 and License = securityk9

R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
R1#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
CISCO1941/K9 platform with 524288 Kbytes of main memory
Main memory is configured to 64/-1(On-board/DIMM0) bit mode with ECC disabled

Readonly ROMMON initialized

program load complete, entry point: 0x80803000, size: 0x1b340
program load complete, entry point: 0x80803000, size: 0x1b340

IOS Image Load Test
___________________
Digitally Signed Release Software
program load complete, entry point: 0x81000000, size: 0x2bb1c58
Self decompressing the image :
########################################################################## [OK]
Smart Init is enabled
smart init is sizing iomem
                  TYPE      MEMORY_REQ
     Onboard devices &
          buffer pools      0x01E8F000
-----------------------------------------------
                TOTAL:      0x01E8F000
Rounded IOMEM up to: 32Mb.
Using 6 percent iomem. [32Mb/512Mb]

              Restricted Rights Legend
Use, duplication, or disclosure by the Government is
subject to restrictions as set forth in subparagraph
(c) of the Commercial Computer Software - Restricted
Rights clause at FAR sec. 52.227-19 and subparagraph
(c) (1) (ii) of the Rights in Technical Data and Computer
Software clause at DFARS sec. 252.227-7013.
```

Licencia R2

```
ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot

R2(config)#: %IOS_LICENSE_IMAGE_APPLICATION-6-LICENSE_LEVEL: Module name = C1900 Next reboot lev
securityk9 and License = securityk9

R2(config)#exit
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#copy run start
Destination filename [startup-config]?
Building configuration...
[OK]
R2#reload
Proceed with reload? [confirm]
System Bootstrap, Version 15.1(4)M4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 2010 by cisco Systems, Inc.
Total memory size = 512 MB - On-board = 512 MB, DIMM0 = 0 MB
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     Onboard devices &
          buffer pools      0x01E8F000
-----------------------------------------------
                TOTAL:      0x01E8F000
Rounded IOMEM up to: 32Mb.
Using 6 percent iomem. [32Mb/512Mb]
```

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3.Implementacion de ACLs

Se crea una lista de acceso que define qué tráfico se va a cifrar.Solo el tráfico entre la red 192.168.1.0 y 192.168.3.0 viajará por el túnel VPN.

R1

```
R1#
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#access-list 110 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
R1(config)#
```

R2

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#access-list 110 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
R2(config)#
```

4.Phase 01: ISAKMP policy

Aquí se establece la conexión segura inicial entre routers.

Se definen método de cifrado, autenticación y la clave compartida (pre-shared key).

R1

```
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes 256
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config-isakmp)#exit
R1(config)#crypto isakmp key secretkey address 209.165.200.1
R1(config)#
```

R2

```
R2(config)#crypto isakmp policy 10
R2(config-isakmp)#encryption aes 256
R2(config-isakmp)#authentication pre-share
R2(config-isakmp)#group 5
R2(config-isakmp)#exit
R2(config)#crypto isakmp key secretkey address 209.165.100.1
R2(config)#
```

5.Phase 02: IPSec transform set.

Se especifica cómo se cifrarán los datos dentro del túnel:

tipo de cifrado (AES) y método de integridad (SHA).

R1

```
R1(config)#crypto ipsec transform-set R1->R2 esp-aes 256 esp-sha-hmac
R1(config)#
```

R2

```
R2(config)#crypto ipsec transform-set R2->R1 esp-aes 256 esp-sha-hmac
R2(config)#
```

6.Crear el mapa criptográfico

R1

```
R1(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R1(config-crypto-map)#set peer 209.165.200.1
R1(config-crypto-map)#set pfs group5
R1(config-crypto-map)#set security-association lifetime seconds 86400
R1(config-crypto-map)#set transform-set R1->R2
R1(config-crypto-map)#match address 100
R1(config-crypto-map)#
```

R2

```
R2(config)#crypto map IPSEC-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
        and a valid access list have been configured.
R2(config-crypto-map)#set peer 209.165.100.1
R2(config-crypto-map)#set pfs group5
R2(config-crypto-map)#set security-association lifetime seconds 86400
R2(config-crypto-map)#set transform-set R2->R1
R2(config-crypto-map)#match address 100
R2(config-crypto-map)#
```

7.Aplicar el mapa criptográfico

El mapa se coloca en la interfaz que da a internet.

R1

```
R1(config)#int g0/0
R1(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R1(config-if)#
```

R2

```
R2(config)#int g0/0
R2(config-if)#crypto map IPSEC-MAP
*Jan  3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
R2(config-if)#
```